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Toehold-mediated DNA displacement-based surface-enhanced Raman scattering DNA sensor utilizing an Au–Ag bimetallic nanodendrite substrate

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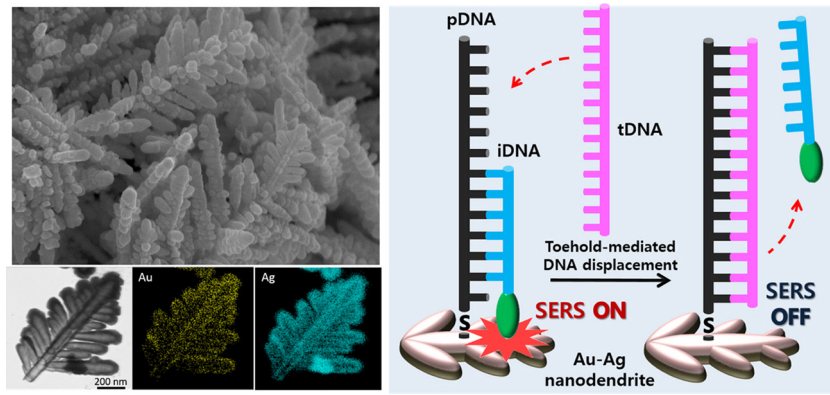
HIGHLIGHTS

- A toehold-mediated DNA displacement-based SERS senor has been demonstrated.
- For DNA sensing, an Au–Ag bimetallic nanodendrite was used as a substrate.
- Ag provided the sensitivity and Au improved the reproducibility of measurement.
- The achieved LOD was 96.3 fM, superior to those reported in previous studies.

GRAPHICAL ABSTRACT



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ABSTRACT

A simple and sensitive surface enhanced Raman scattering (SERS)-based DNA sensor that utilizes the toehold-mediated DNA displacement reaction as a target-capturing scheme has been demonstrated. For a SERS substrate, Au–Ag bimetallic nanodendrites were electrochemically synthesized and used as a sensor platform. The incorporation of both Ag and Au was employed to simultaneously secure high sensitivity and stability of the substrate. An optimal composition of Ag and Au that satisfied these needs was determined. A double-strand composed of 'a probe DNA (pDNA)' complementary to 'a target DNA (tDNA)' and 'an indicator DNA tagged with a Raman reporter (iDNA)' was conjugated on the substrate. The conjugation made the reporter molecule close to the surface and induced generation of the Raman signal. The tDNA released the pre-hybridized iDNA from the pDNA via toehold-mediated displacement, and the displacement of the iDNA resulted in the decrease of Raman intensity. The variation of percent intensity change was sensitive and linear in the concentration range from 200 fM to 20 nM, and the achieved limit of detection (LOD) was 96.3 fM, superior to those reported in previous studies that adopted different signal taggings based on such as fluorescence and electrochemistry.

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1. Introduction

DNA sensors utilizing single-strand DNA probes have been widely studied for selective detection of complementary DNA targets. However, the detection of a target DNA normally requires time- and labor-consuming steps of repeating incubation and washing at elevated temperatures to minimize any mismatched hybridization that could deteriorate the selectivity of the measurement [1–3]. Recently, the DNA strand displacement reaction, which deploys a pre-formed double-strand probe (dsProbe) system composed of an unlabeled capture strand and a labeled reporter strand, was introduced as a simpler and more effective scheme for DNA detection [4–6]. When a target presents, the reporter strand is forced to come away from the double-strand and the resulting signal change by the displacement is used for sensing.

Toehold-mediated DNA strand displacement reactions, employing an unhybridized base region (named the toehold) at end of a duplex (double strand) to trigger DNA rehybridization, have drawn significant attention. The toehold mediation provides a thermodynamic driving force for hybridization and accelerates strand displacement even at room temperature [6–10]. Therefore, the displacement of a pre-hybridized reporter strand could be more efficient using the toehold mediation compared with other displacement methods such as enzymatic and central mismatched displacements. Due to these advantages, several biosensors based on toehold-mediated DNA displacement have been recently reported. Larkey et al. [11] demonstrated a miRNA fluorescence sensor utilizing fluorescence quenching that occurs when two different dyes tagged at both ends of a freed reporter DNA became close by forming a hairpin structure (limit of detection (LOD): 10 nM). A quartz crystal microbalance (QCM)-based sensing scheme was also studied [12]. A hairpin capture probe became an open-loop structure by hybridization with a target DNA and then a streptavidin coupled reporter strand was allowed to bind on the opened probe for mass amplification (LOD: 0.3 nM). Alternatively, two different electrochemical DNA (E-DNA) sensors using methylene blue as a redox reporter are presented [13,14]. In these sensors, the capture of a target DNA allows the reporter to approach the electrode surface and increases current through enhanced electron transfer (LODs: 50 pM and 400 fM).

Meanwhile, surface enhanced Raman scattering (SERS) is highly useful for sensitive analytical sensing due to the significant enhancement of the Raman signal by resonance with local plasmon generated on the SERS substrate, thus it has been widely adopted for many applications that demand sensitivity [15–18]. Therefore, the combination of both merits from SERS and the toehold-mediated displacement reaction for DNA detection could be synergetic. Development of a microfluidic system sequentially running the toehold-mediated displacement reaction and SERS detection of released DNAs was attempted [19]. Silica beads pre-immobilized with dsProbe were mixed with target DNA (tDNA) to initiate the displacement reaction and the displaced DNA strand possessing a Raman reporter then bound to prepared bare Ag nanoparticles (AgNPs) in a fluidic channel for generation of the SERS signal. The detection level was 100 pM, comparable to those achieved in the previously mentioned investigations; while, a simpler sensing scheme rather than two separate steps of DNA displacement and conjugation of the released DNA on AgNPs, and the improvement of sensitivity incorporating more SERS-efficient substrates is further beneficial.

Here, we developed an Au–Ag bimetallic nanodendrite possessing large numbers of hot spots within the structure and used as a SERS substrate to enhance detection sensitivity. The purpose of the bimetallic composition was to harmonize individual advantages of Au for stability and Ag for sensitivity to maximize sensing performance, and an optimal composition satisfying both demands was determined. To measure a tDNA, a dsProbe composed of a probe DNA (pDNA) and an indicator DNA possessing a Raman reporter (iDNA) was initially immobilized on the substrate as shown in Fig. 1. The conjugation of the dsProbe allowed the Raman reporter to associate closely near the substrate surface and turned the Raman signal on. When the tDNA displaced the pre-hybridized iDNA, a decrease of Raman intensity is expected according to the increase of tDNA concentration. Since the magnitude of the SERS signal could vary depending on the distance between the Raman reporter and the surface, an optimal hybridization location of the iDNA on the pDNA was also examined.

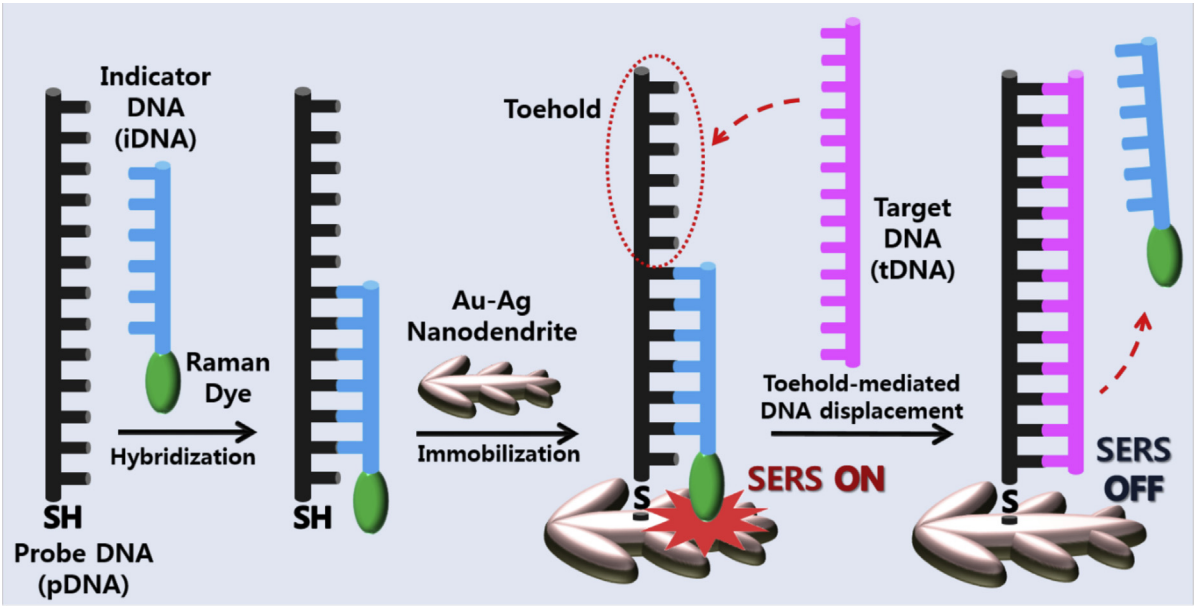

Fig. 1. Schematic presentation of overall sensing procedure for detection of tDNA using toehold-mediated DNA displacement on Au–Ag nanodendrite.

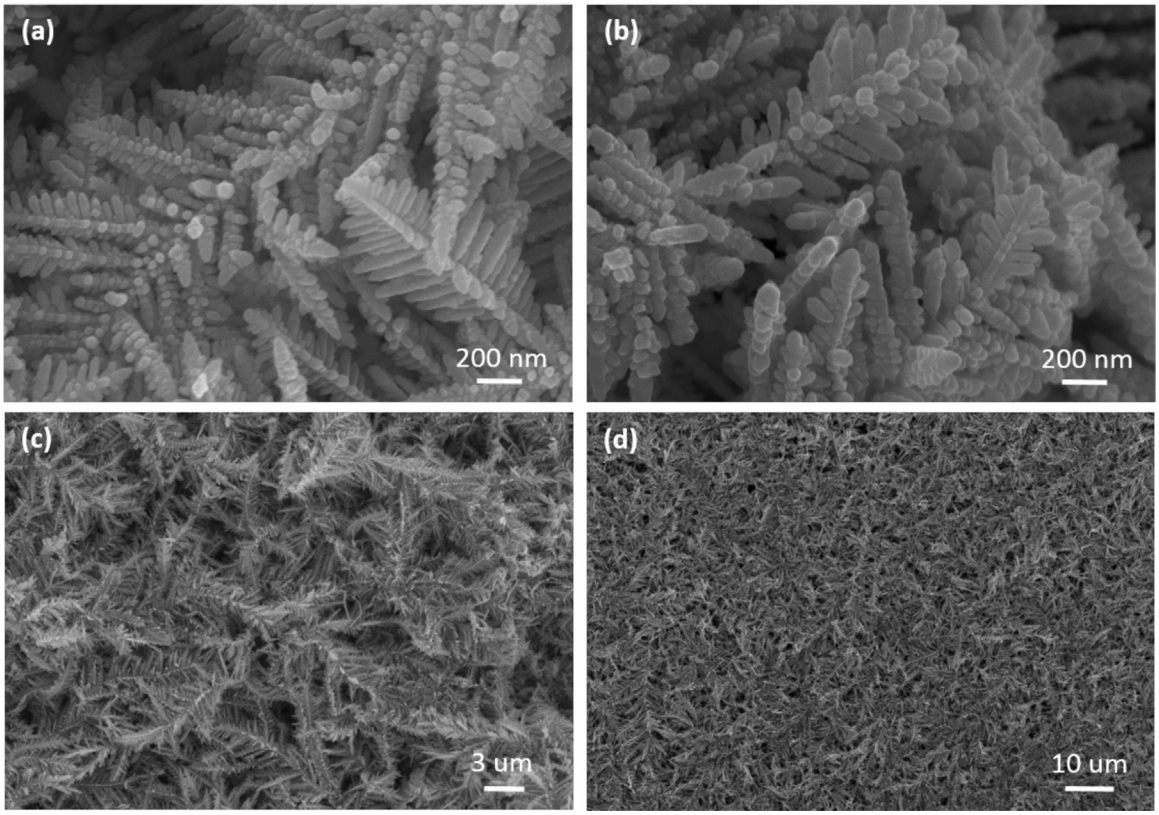

Fig. 2. SEM images ($2.52 \times 1.76\ \mu m^2$) of Ag nanodendrite (a) and $Au_{3.17}$–$Ag_{96.83}$ bimetallic nanodendrite (b). The $Au_{3.17}$–$Ag_{96.83}$ nanodendrite images covering larger areas of 42.4×29.6 (c) and $116.5 \times 81.6\ \mu m^2$ (d) are also shown.

Finally, different concentrations of tDNA samples were measured at the optimized sensor conditions, and the resulting analytical performances, including reproducibility and LOD, were evaluated.

2. Experimental

2.1. Electrochemical synthesis of Au–Ag nanodendrites

All necessary chemicals for synthesis, including $AgNO_3$ (99.9%), $HAuCl_4$ (99.999%), citric acid, were purchased from Sigma–Aldrich. For the preparation of Ag nanodendrite, a Ti foil (1×2 cm with a thickness of 0.032 cm, 99.7%, Alfa Aesar) was immersed in a 2 mL solution composed of 10 mM $AgNO_3$ and 1 M citric acid. Then, using a galvanostat (Epsilon, Bioanalytical Chemistry Systems, Inc., USA), a current of 450 μA was applied to the foil for electrodeposition of Ag over 13 min with a Pt counter and the Ag/AgCl reference electrode [20]. After the electrodeposition, the substrate was carefully washed with deionized (DI) water. Next, to construct Au–Ag bimetallic nanodendrites, the prepared Ag nanodendrite was dipped in a 3 mL solution of 0.1 mM $AuCl_4^-$ and a galvanic replacement reaction (GRR) incorporating Au by releasing Ag in the nanodendrite was executed with continuous stirring. The duration of GRR was varied from 1 to 12 min to prepare Au–Ag nanodendrites with different atomic compositions. Potentially adsorbed AgCl, a product of GRR, on the substrates was cleaned by washing several times using a saturated NaCl solution. Scanning electron microscope (SEM) images and energy dispersive X-ray (EDX) spectra of the substrates were acquired using a Hitachi S-4800 SEM instrument. Transmission electron microscope (TEM) and atomic mapping images were obtained using a JEM-2100F TEM (JEOL, Japan) with an accelerating voltage of 200 kV.

2.2. SERS measurement and DNA displacement reaction

To find an optimal Au–Ag nanodendrite providing maximal SERS intensity with good reproducibility, synthesized nanodendrites of different atomic compositions were immersed in a 600 μL 2 μM 2-naphthalenethiol (2-NpSH) solution over 4 h under dark environment. Then, Raman spectra were acquired on the 2-NpSH adsorbed nanodendrites using a dispersive Raman microscope (Kaiser Optical Inc., USA) equipped with a diode laser (λ_{ex} = 785 nm), a thermoelectrically cooled (−40 °C) charge-coupled device (CCD) detector, and a holographic grating (resolution: 4 cm^{-1}). In each case, spectra were acquired at 15 randomly selected spots on each substrate and average spectrum of them was used for optimization of sensing conditions and quantitative analysis.

The pDNA (5′HS-$(CH_2)_6$-AGGCCCGGGAACGTATTCCA), iDNA (5′CGTTCCCGGGCC-Cy5) and tDNA (5′TGGAATACGTTCCCGGGCCT) were purchased from Bioneer Inc. (Daejeon, Korea) [21]. Hexanethiol (HS) at the terminal of the pDNA was used for its immobilization on nanodendrite surface. Cy5 (Cyanine 5), a Raman reporter molecule, was linked to the iDNA to generate a Raman signal as shown in Fig. 1. Equal amounts of the pDNAs and iDNAs were mixed and hybridized to form the double strand (duplex) in a 10 mM of pH 7.2 phosphate buffer containing 0.3 M NaCl, and then 500 nM of the diluted duplex was allowed to bind on the nanodendrite surface overnight. To prevent non-specific binding, the duplex bound nanodendrite was allowed in a 1 μM mercaptohexanol solution to passivate the empty surface over 1 h and then was carefully washed with DI water for sensing of the tDNA. Six different concentrations (200 fM–20 nM) of tDNA samples were prepared in a 10 mM phosphate buffer with 150 mM NaCl and the nanodendrite sensors were positioned in the samples to initiate a

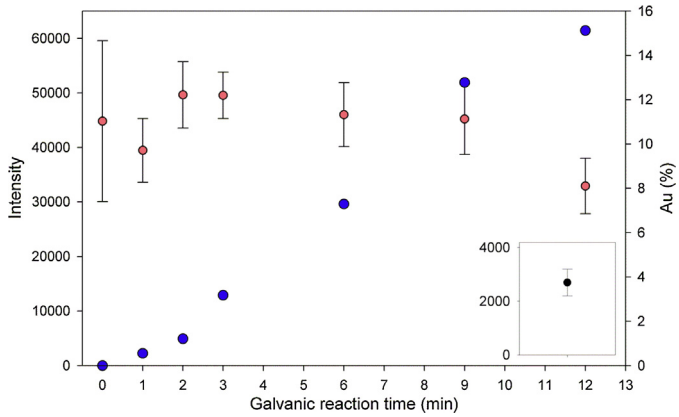

Fig. 3. Variation of Au concentration in bimetallic nanodendrites with the increase of GRR time (blue colored circles and right *y*-scale) and the corresponding intensities of the adsorbed 2-NpSH peak at 1378 cm^{-1} (red colored circles, left *y*-scale). The intensity acquired using the Au nanodendrite is shown in the inset for comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

toehold mediated DNA displacement reaction. After completion of the reaction, the sensor was washed with buffer and DI water, and a laser was illuminated on the surface to acquire the Raman spectra. The percent signal change measured before and after the execution of the displacement reaction was used to correlate the concentration variation of tDNA.

3. Results and discussion

3.1. Determination of Au and Ag concentrations in a nanodendrite for optimal SERS measurement

Fig. 2(a) shows a SEM image of Ag nanodendrite constructed by electrodeposition ($2.52 \times 1.76\ \mu m^2$). Nanodendrites of a hierarchical structure with a high population of edges as well as acute angles in terraces are clearly observed. The fabrication of bimetallic Au–Ag was accomplished by GRR since the standard redox potential of Ag^+/Ag (0.8 V vs. SHE) was lower than that of Au^{3+}/Au (1.5 V), so the incorporation of Au into the pre-prepared Ag nanodendrite was thermodynamically favorable. By increasing the duration of GRR up to 12 min, six Au–Ag nanodendrites with different compositions were synthesized and their atomic compositions were determined using EDX analysis by calculating ratios of the corresponding peaks. Fig. 3 shows the variation of the Au concentration according to the increase of GRR time (blue colored circles and right *y*-scale). The concentration of Au obviously increased with the longer GRR and reached up to 15.1% by a GRR of over 12 min.

Among the prepared Au–Ag nanodendrites, a substrate providing the highest SERS signal was investigated. For this purpose, the substrates were allowed in a 2 mM 2-NpSH solution for 4 h and Raman intensities of adsorbed 2-NpSH at 1378 cm^{-1} (the strongest peak) [22] were examined as shown in Fig. 3 (red colored circles, left *y*-scale). Before calculation of the intensities, the baselines of the spectra were corrected at 1408 and 1306 cm^{-1}. At a given GRR time, 3 nanodendrites were independently prepared and an average spectrum of 15 spectra collected by randomly illuminating laser on different spots from each substrate was used for evaluation. The error bars correspond to standard deviations of intensities measured from three separate substrates in each case. For comparison, an Au nanodendrite was also prepared as described in a previous publication [23] and the same measurement was performed as shown in the inset.

When the intensities acquired from the nanodendrites solely composed of either Ag or Au are compared, the Ag nanodendrite yields a 17 times higher SERS signal since Ag is a more efficient SERS enhancer than Au as known. Meanwhile, the standard deviation of intensities measured from the Ag nanodendrite is high with a relative standard deviation (RSD) of 34.6%, since the Ag surface is quite unstable at ambient conditions due to the easy formation of an oxide layer. The RSD yielded from the Au nanodendrite was 9.2%, indicative of more stable surface. Therefore, the partial incorporation of stable Au into Ag nanodendrite could improve the surface stability, which is directly related with the reproducibility of the SERS measurement. One noticeable observation is that the intensity fluctuation (standard deviation) started to decrease with the Au incorporation. In the case of the $Au_{3.17}$–$Ag_{96.83}$ nanodendrite (3.17% Au and 96.83% Ag), the resulting RSD was 8.7%, approximately a 4-fold improvement relative to that obtained from the Ag nanodendrite. The magnitude of intensity gradually decreases when the GRR time is longer than

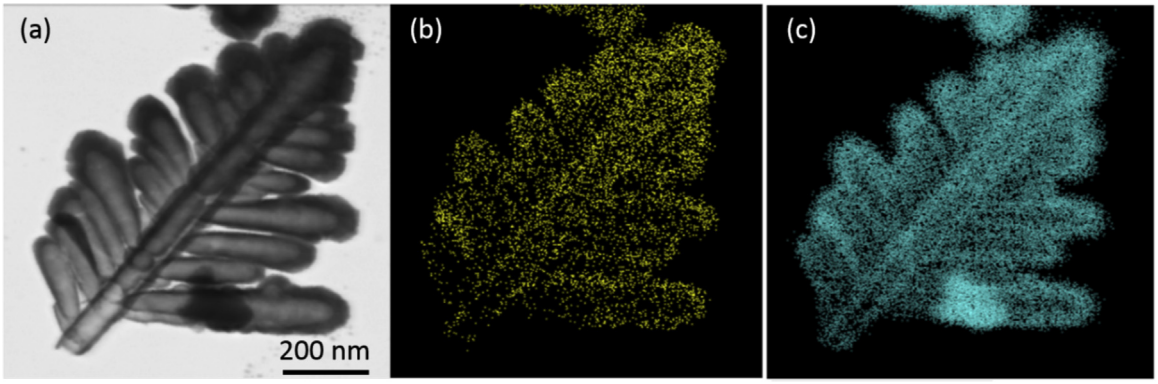

Fig. 4. TEM image of single $Au_{3.17}$–$Ag_{96.83}$ nanodendrite (a) and the corresponding elemental mapping images of Au (b) and Ag (c).

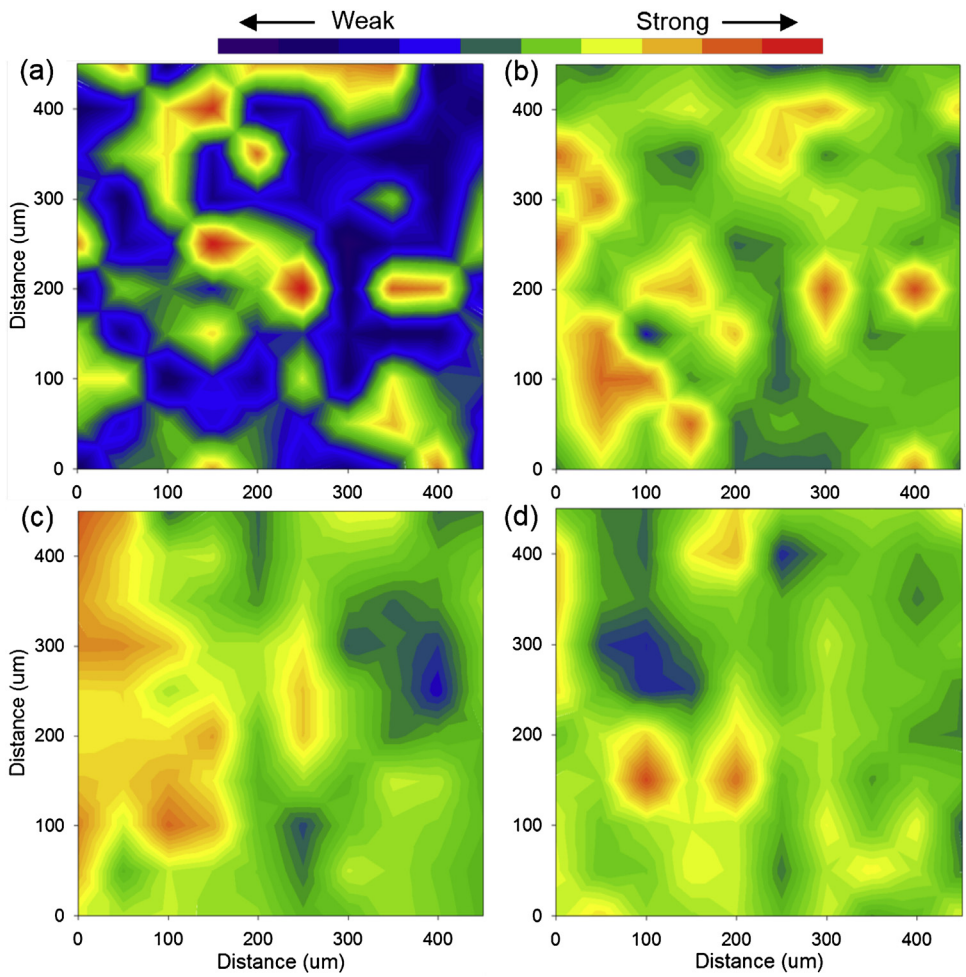

Fig. 5. Raman mapping images of Ag (a) and three different $Au_{3.17}$–$Ag_{96.83}$ nanodendrites ((b), (c), (d)) constructed using the intensity of the 1378 cm^{-1} band. Each image covers the area of $450 \times 450\,\mu m^2$ on the substrate. The red/orange and purple/blue colors indicate the higher and lower SERS intensities, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3 min due to the presence of a larger amount of Au, a less SERS-efficient metal.

Fig. 2(b) shows an SEM image of the $Au_{3.17}$–$Ag_{96.83}$ nanodendrite ($2.52 \times 1.76\,\mu m^2$). In comparison with the pure Ag nanodendrite, the diameter of the dendrite tip increases slightly due to the nucleation of Au randomly on the Ag surface and the formation of layers of Au–Ag alloys as also observed in a previous publication [24], in which the growth dynamics was extensively investigated. To alternatively confirm the Au–Ag alloy formation, a TEM image of a single $Au_{3.17}$–$Ag_{96.83}$ nanodendrite (a) was obtained and the corresponding elemental mapping images of Au (b) and Ag (c) were separately examined as shown in Fig. 4. As shown in the TEM image, a dendritic structure is apparent without serious structural deformation. In both Au and Ag mapping, the corresponding color is uniformly distributed over the structure, thereby confirming the homogeneous Au–Ag alloy formation over the nanodendrite. Although Au does not fully cover the whole structure, the partial incorporation of Au is obviously advantageous to improve the reproducibility as shown in Fig. 3.

The uniformity of a substrate is another important issue to address for reliable SERS-based quantitative analysis. Therefore, SEM images of the $Au_{3.17}$–$Ag_{96.83}$ nanodendrite covering larger areas were initially examined as shown in Fig. 2(c) and (d), corresponding to the field of view of 42.4×29.6 and $116.5 \times 81.6\,\mu m^2$, respectively. As seen in both images, the spread of nanodendrites over the areas are fairly consistent without any noticeable localization. For further investigation, Raman mapping covering the area of $450 \times 450\,\mu m^2$ on the substrate was performed with a spectral collection interval of 50 μm in both the *x* and *y* directions. The mapping was separately executed for three $Au_{3.17}$–$Ag_{96.83}$ nanodendrites as well as Ag nanodendrite as a comparison. A total of 100 spectra were collected in each mapping. Fig. 5 shows Raman mapping images of Ag (a) and thee different $Au_{3.17}$–$Ag_{96.83}$ nanodendrites ((b), (c), (d)) constructed using the intensity of the 1378 cm^{-1} band. In the images of $Au_{3.17}$–$Ag_{96.83}$ nanodendrites, the spots of high intensity (red/orange color) are widely spread over the mapped areas without distinctly intensity-localized spots; while, the intensities largely fluctuate in the mapping of Ag nanodendrite. RSDs of the mapped intensities (100 observations) in Fig. 5(a), (b), (c) and (d) were 30.7, 9.7, 8.3 and 7.2%, respectively.

Since Raman intensities of adsorbed 2-NpSH on the Au–Ag nanodendrite still varies depends on the locations as shown above, it is necessary to confirm whether an average spectrum of 15 randomly collected spectra is quantitatively representative of SERS measurement for a given substrate or not. For this purpose, 15 spectra were randomly selected out of 100 mapped spectra and RSD of the intensities at 1378 cm^{-1} in the chosen spectra was calculated. This procedure was repeated 200 times and the resulting 200 RSDs were averaged for the evaluation. The average RSDs for the cases in Fig. 5(b), (c) and (d) were 9.5, 8.7 and 7.1%, respectively, nearly similar to those acquired using all 100 mapped

spectra. Therefore, the use of a 15 spot-averaged spectrum for a substrate was proper to represent SERS intensity for subsequent analysis. Although the number of spectra for average increased more than 15, there was no significant variation in the RSD of the intensities. Hereinafter, the $Au_{3.17}$–$Ag_{96.83}$ nanodendrite was solely employed as a SERS sensor platform for detection of tDNAs and the average intensity based on 15 random spectral acquisitions on each substrate was used for the evaluation of sensing performances.

3.2. Optimization of hybridization location of iDNA and displacement reaction time

As shown in Fig. 6(a), the distance between Cy5 (Raman reporter) and the nanodendrite surface depends on where the iDNA is hybridized on the pDNA, and Raman intensity could vary according to the hybridized location of iDNA. Therefore, it was necessary to determine an optimal hybridization location of the iDNA that could provide maximal Raman signal for sensitivity of measurement. To accomplish this, three iDNAs (12mer) with different base sequences designed to bind on different locations of the pDNA were prepared as described in the figure. Case I corresponds to the hybridization of iDNA on the third through fourteenth bases of pDNA. Cases II and III are to initiate hybridization of iDNA from the second and the first base on the pDNA, respectively. Therefore, the distance between Cy5 and the surface is longest and shortest in case I and case III, respectively. The molecular structures of 1-hexanethiol and Cy5 with a linker are also shown in the figure. The concentration of the duplex before the immobilization was 500 nM in each case.

Fig. 6(b) shows Raman spectra of Cy5 acquired at cases I (red), II (green) and III (blue). The baselines of raw spectra were corrected at 1540, 1423, 1339, 1250, 972, 822, 600 and 450 cm^{-1}. The band positions observed here are equivalent to those of the same Cy5 SERS spectrum reported in a previous publication [25]. The 1540–1425 cm^{-1} range including the most intense band at 1468 cm^{-1} is highlighted in the left inset. In addition, the intensities of 1468 cm^{-1} band acquired in cases I, II and III are shown in the right inset. Three separately prepared nanodendrites were employed for the evaluation in each case. Case II provides the strongest Raman intensity, although case III makes the distance between Cy5 and the surface shortest. This may be because the size of Cy5 is bigger than that of 1-hexanethiol and the linker chain including an amide bond is additionally contained in Cy5. The size of the 1-hexanethiol part in the probe DNA is 1.2 nm as previously reported [26] and the base-to-base distance in DNA is 0.32 nm as generally known. Therefore, the available space provide by case III

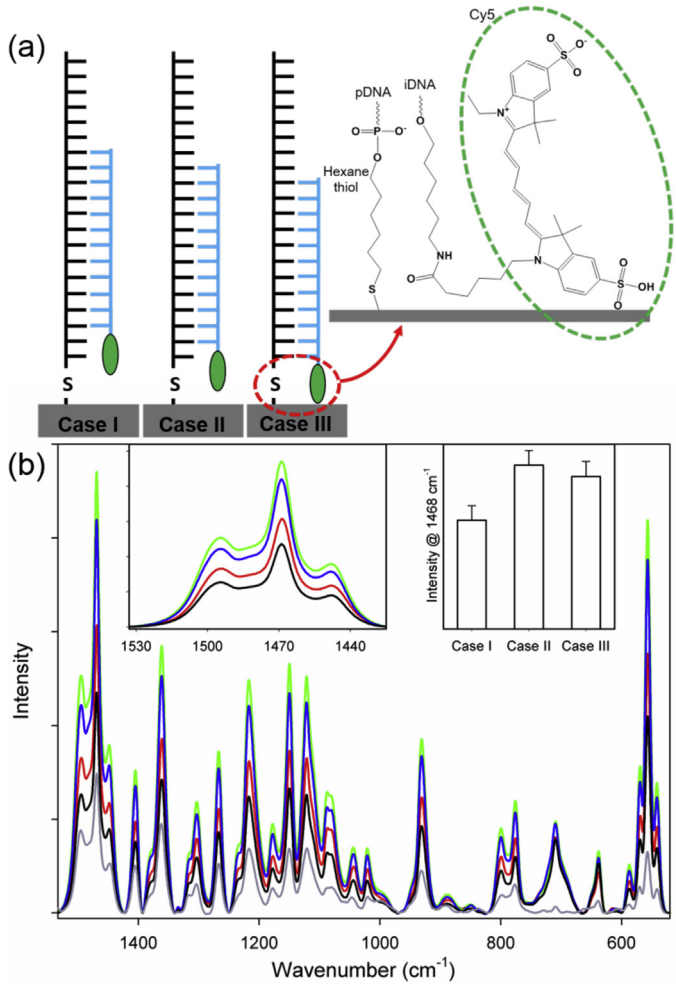

Fig. 6. (a) Graphical description for three different cases (I, II and III) of hybridizing the iDNA on the pDNA. The molecular structures of 1-hexanethiol and Cy5 with a linker are also shown. (b) Baseline-corrected Raman spectra of Cy5 acquired for case I (red), II (green) and III (blue). Raman spectrum acquired at 60 min of the displacement reaction on case II executed with 200 pM tDNA sample (black) is also shown. The left and right inset shows the 1540–1425 cm^{-1} range including the most intense band at 1468 cm^{-1} and the intensities of 1468 cm^{-1} band in case I, II and III acquired from three separately prepared nanodendrites, respectively. The grey spectrum corresponds to normal Raman of 7 mM Cy5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

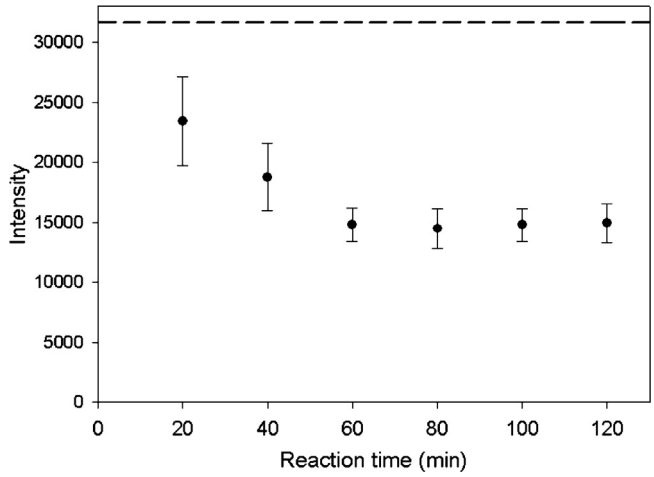

Fig. 7. The intensity variation of the 1468 cm^{-1} band of Cy5 as the displacement reaction proceeds up to 120 min using a 200 pM tDNA sample. The dashed line indicates the band intensity before the addition of sample.

could be limited to fully occupy the Cy5 unit and a certain degree of structural distortion, which would decrease the efficiency of SERS, could occur. Case II provides the increased distance by 0.32 nm, so the occupation of Cy5 unit is more facile and proper to generate SERS signal efficiently. When the distance increases further (case I), the intensities of Raman bands fairly decreases since Cy5 becomes further away from the influence of the hot spot. RSDs of the intensities in cases I, II and III were 13.1, 9.0 and 9.9%, respectively. Case II also affords the more reproducible measurement.

Furthermore, the orientation of the immobilized duplex composed of pDNA and iDNA on nanodendrite surface is not always straight up as depicted in the figure [27]. In reality, it is expected that the orientation varies greatly. Consequently, the vertical distance from the terminal of the iDNA to the surface changes accordingly and the orientation of Cy5 is hardly predictable; therefore, the optimal hybridization location of iDNA needs to determine experimentally as performed in this study. Finally, the iDNA was hybridized on the second through thirteenth bases on the pDNA (case II) for subsequent sensing of tDNAs. The maximized Raman signal at this condition is reasonable when the sizes of 1-hexanethiol and Cy5 unit are simply compared. The grey spectrum corresponds to normal Raman of 7 mM Cy5. Both SERS and normal Raman spectral feature are generally similar to each other. This result indicates that the enhancement of Raman intensity is not specific to certain vibrations since the orientation of Cy5 on the substrate surface is not isotropic as described above, while the 709, 638 and 557 cm^{-1} bands are relatively more enhanced with SERS measurement.

An optimal duration for the toehold-mediated displacement reaction was also determined for reliable measurement. A sample of 200 pM tDNA was used for the evaluation by changing the displacement reaction time up to 120 min and resulting intensity variation was examined. Fig. 7 shows the intensity variation of the 1468 cm^{-1} band (the most intense peak) as the displacement reaction proceeds up to 120 min. The dashed line indicates the band intensity before the addition of the tDNA. A decrease of the intensity was observed with the progress of displacement reaction that forces the iDNA to leave from the pre-existing duplex. The intensity decrease continued up to 60 min and no meaningful intensity change occurred afterward. This observation is analogous with those reported in previous publications [12,19] in which a double-strand was also pre-immobilized on a substrate surface before measurements; while, the displacement was completed approximately in 10 min when a double-strand was in a solution without immobilization on the surface [7,11,28]. The Raman spectrum acquired at 60 min of the displacement from the 200 pM sample (black) is also shown in Fig. 6(b). The intensity decreased significantly by the releases of the iDNA and the percentage of intensity decrease is 52.6%. In the long run, the sensors were prepared by hybridizing the iDNA according to the

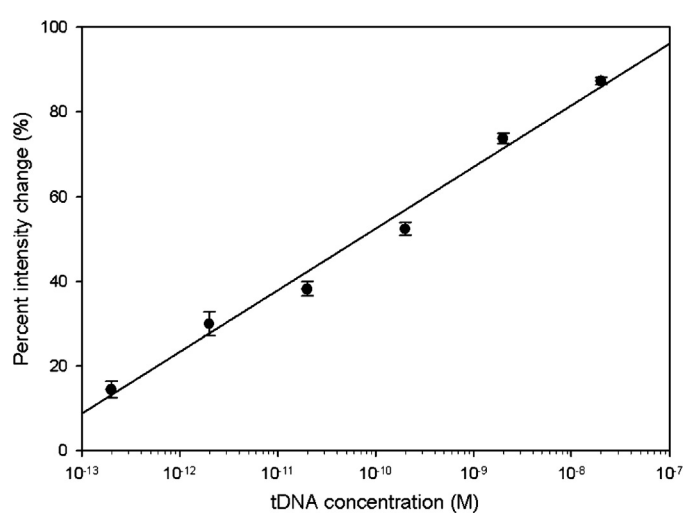

Fig. 8. Concentration correlation between tDNA concentrations in a logarithm scale and percent intensity changes of corresponding Raman signals.

case II scheme and the displacement reaction was accomplished over 60 min for the detection of tDNAs in various concentrations.

3.3. Quantitative determination of tDNAs in different concentrations

Six tDNA samples with a concentration range from 200 fM to 20 nM were measured using the Au–Ag nanodendrite SERS sensor at the optimized conditions. Fig. 8 shows the percent intensity change measured from each sample. At each concentration, both Raman signals before and after the addition of sample were obtained and the percent intensity change was used to correlate with the concentration variation. As performed earlier, three independently prepared nanodendrites were also used for the measurement. The percent intensity change becomes greater in a linear fashion ($R^2 = 0.985$) with the increase of tDNA concentration in a logarithm scale. Increased amounts of tDNA make the pre-hybridized iDNA more displaced, so a larger decrease of the intensity results. It was noticed that the magnitudes of the error bars were small (average RSD: 6.3%) since the measurement is based on a comparison of two separate signals obtained from the same substrate. The signal acquired before the displacement was a good reference to compensate for possible substrate-to-substrate fluctuation of the SERS signal. The LOD calculated according to the equation ($LOD = LOB + 1.645(SD_{low\text{-}concentration\text{-}sample}$, where LOB (Limit of Blank) $= mean_{blank} + 1.645(SD_{blank})$) was 96.3 fM [29], superior to those in the reports described in the introduction.

4. Conclusion

We have demonstrated a novel SERS DNA sensor in conjunction with a bimetallic nanodendrite structure as a sensitive and efficient SERS platform for the toehold-mediated DNA displacement reaction. The achieved LOD was superior (96.3 fM) and the sensor-to-sensor reproducibility was good. Furthermore, we believe that our DNA sensor is clinically applicable and can used for detecting a range of DNA samples, and in particular will be of value for diagnosing inherited diseases such as Avellino corneal dystrophy that originate from point mutation. We are under developing a new analytical method for identification of the disease using a more efficient SERS system using trimetallic nanostructures and target recognizing schemes with improved specificity.

Acknowledgements

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